

Course: T.Y.B. Tech.

Semester: VI

Subject Code & Name: Digital System Design (BTETOE605A)

Max Marks: 60

Date: 20/08/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Calculators is not allowed.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Convert the following expression in standard SOP form $F=A+B'C$ 06
- B) Implement full adder and full subtractor using decoder and OR gates. 06
- C) What is State diagram? Explain in detail state table and excitation tables with suitable example 06

Q.2 Solve Any Two of the following.

- A) Derive characteristic equations for J-K, D and T F/Fs 06
- B) Draw Universal shift Register with following operations. 06

Control or select line	Operation
11	Parallel load
10	Shift left
01	Shift right
00	No operation

- C) What is difference between TTL and CMOS. 06

Q.3 Solve Any Two of the following.

- A) Solve the following example using 4-variable K-map $F=\sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$ 06
- B) Design logic circuit to convert 3-bit Binary code into a Gray code 06
- C) Draw clock waveform (timing diagram) for BCD twisted counter. 06

Q.4 Solve Any Two of the following.

- A) Design 3 bit synchronous sequential counter using T flipflop. 06
- B) Define following terms. 06
- | | | |
|----------------------|---------------|----------------|
| i) Propagation delay | ii) rise time | iii) Fall time |
| iv) delay time | v) fan-in | vi) fan-out |
- C) Implement 5 to 32 decoder using only 3 to 8 decoders. 06

Q.5 Solve Any Two of the following.

- A) How many types of memory are there in digital electronics? 06
- B) Write VHDL code for 4x1 MUX using dataflow and behavioral architecture style. 06
- C) Explain FPGA architecture with suitable diagram. 06

*** End ***